
Disentangling Representation from Causation for the AI-Style Direction in Transformer Residual Streams

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Abstract

1 Large language models (LLMs) produce outputs with a recognizable stylistic sign-
2 nature that humans readily identify as AI-generated. Prior work has demonstrated
3 that transformer language models encode the property of "sounding like AI" as a
4 near-linear direction in the residual stream, extractable via Contrastive Activation
5 Analysis (CAA); however, these findings are substantially confounded by text
6 length [Hypogenic-AI, 2024]. In this work, we investigate whether this direction is
7 genuine beyond verbosity and, critically, whether the layer from which it is most
8 cleanly extracted is also causally operative during generation. Using CAA on a
9 length-matched version of the HC3 dataset, we extract a genuine AI-style direction
10 classifiable at 96.5–98.0% accuracy on both QWEN2.5-3B and QWEN2.5-3B-
11 INSTRUCT. We then find a sharp dissociation between representation and causation:
12 the layer that most linearly encodes the extracted direction is causally inert, while a
13 distinct region in the middle of the network drives stylistic generation. Surgical ab-
14 lation of this region meaningfully reduces AI-likeness but at a tradeoff with writing
15 coherence, with QWEN2.5-3B-INSTRUCT consistently achieving a more favorable
16 style-coherence balance than QWEN2.5-3B. We additionally propose and evaluate
17 a detector-weighted variant of CAA, finding that re-weighting training examples by
18 external detector confidence does not improve ablation effectiveness. More broadly,
19 our work showcases how an understanding of model internals can be leveraged to
20 develop more robust AI-detection and AI-style intervention techniques.

21 1 Introduction

22 LLM outputs have a distinctive stylistic signature that humans readily recognize. They often have
23 minimal hedged phrasing, a more formal register, and bullet-pointed structuring. AI text detection
24 systems increasingly inform policy decisions in academic integrity, content moderation, and synthetic
25 media governance, yet the mechanistic encoding of AI-style within transformer models remains
26 poorly understood. If the model itself represents AI-style as a structured internal direction, this opens
27 principled paths to both more robust detection and more natural generation. Inspired by the rapid
28 progress of mechanistic interpretability, this work leverages the internal representations of LLMs to
29 better understand how AI-style is encoded.

30 It is widely hypothesized that LLMs represent high-level semantic features, or concepts, as lin-
31 ear directions in residual stream activations [Park et al., 2023]. A growing body of work has
32 extracted causal linear directions for specific behaviors: truthfulness [Marks and Tegmark, 2024],
33 sentiment [Konen et al., 2024], refusal Arditi et al. [2024], and sycophancy [Panickssery et al., 2024].
34 Moreover, these feature directions have been shown to be effective casual mediators of behavior,
35 enabling fine-grained steering of model outputs with minimal capability cost. Thus, this work aims to

36 provide insight on whether "sounding like AI" is linearly encoded in the residual stream, and whether
 37 that encoding is causally operative during generation.

38 Prior work has demonstrated that transformer language models encode the property of "sounding like
 39 AI" as a near-linear direction in the residual stream, classifying AI versus human text with 97.5%
 40 accuracy via CAA on the HC3 dataset. However, these findings are substantially confounded by
 41 text length with a cosine similarity of 0.93 between the "AI direction" and a length-based direction
 42 [Hypogenic-AI, 2024]. In this work, we show that a near-linear AI-style direction beyond verbosity
 43 exists in the residual stream of QWEN2.5-3B and QWEN2.5-3B-INSTRUCT using a length-matched
 44 version of the HC3 dataset. We additionally propose a detector-weighted variant of CAA that
 45 re-weights training examples by an external detector’s confidence score, with the intuition that
 46 prototypically AI-like examples should contribute more to the extracted direction. We then investigate
 47 whether this direction is causally operative during generation through a systematic layer-window
 48 ablation sweep across all 36 layers of the network.

49 Our central finding is a sharp *representation-causation dissociation*: the layer that most linearly
 50 separates AI from human text activations is causally inert as ablating it during generation produces
 51 near-zero change in measured AI-likeness. Instead, a causal generation region in middle layers
 52 15-22 is responsible for stylistic output. A 5-layer window ablation centered on this region reduces
 53 LLM-judge AI-likeness meaningfully, but always at some tradeoff with writing coherence. Instruction
 54 fine-tuning further separates style from capability: the instruct model shows a more favorable style-
 55 quality tradeoff under surgical ablation, consistent with RLHF producing a more factorized residual
 56 stream. These findings suggest that linear probing accuracy is not a reliable indicator of causal
 57 relevance for AI-style.

58 We make the following **contributions**:

- 59 • We construct a length-matched variant of HC3 and extract a genuine AI-style direction achieving
 60 96.5–98.0% accuracy on both QWEN2.5-3B and QWEN2.5-3B-INSTRUCT, with the verbosity
 61 confound explicitly eliminated (length cosine similarity drops from 0.93 to -0.065).
- 62 • Via a systematic sliding-window layer sweep ($k \in \{1, 5\}$ across all 36 layers), we identify a
 63 representation-causation dissociation: the best-classifying layer (28–29) is causally inert during
 64 generation, while middle layers 15–22 constitute the true causal generation region.
- 65 • We demonstrate that a 5-layer window ablation centered on the causal region reduces LLM-judge
 66 AI-likeness by ~ 3 points on a 7-point scale, always at some tradeoff with writing coherence. The
 67 instruct model exhibits a consistently more favorable style-coherence tradeoff than the base model
 68 across all conditions, consistent with RLHF producing a more factorized residual stream.
- 69 • We show that detector-weighted CAA using three detectors with distinct decision boundaries
 70 (RoBERTa, RADAR, GPTZero) does not improve upon standard CAA in ablation effectiveness,
 71 suggesting that external detectors do not capture a more causally concentrated stylistic signal than
 72 the unweighted mean difference.

73 2 Methodology

74 2.1 Dataset Construction

75 We use HC3 [Guo et al., 2023], a corpus of about 40K paired human and ChatGPT responses drawn
 76 primarily from the ELI5 subreddit and several question-answering sources. The original dataset
 77 exhibits a systematic length asymmetry: AI responses average approximately 914 characters versus
 78 446 for human responses with standard deviations of 340 and 279 characters respectively. An AI-style
 79 CAA direction extracted from HC3 will therefore be significantly confounded by verbosity.

80 To control for the verbosity confound, we construct a *length-matched* variant using a sentence-
 81 boundary-aware truncation algorithm. For each (human, AI) pair, we identify the shorter response
 82 and truncate the longer to the same character count, snapping to the nearest sentence boundary to
 83 avoid mid-sentence cuts. Pairs with a length ratio exceeding 5.0 or a post-truncation length below 50
 84 characters are discarded. The resulting dataset has 30,758 (human, AI) pairs with mean lengths of
 85 approximately 519 characters for human responses and 468 characters for AI responses with standard
 86 deviations of 277 and 295 respectively. We use a 200/50/100 train/validation/test split matching prior
 87 work on this dataset for direct comparability.

88 2.2 Models

89 We study two variants from the Qwen2.5 family [Qwen et al., 2025]. QWEN2.5-3B is a 3B-
 90 parameter decoder-only transformer with 36 layers and a hidden dimension of 2048, trained on a large
 91 multilingual pretraining corpus. QWEN2.5-3B-INSTRUCT is the corresponding instruction-tuned
 92 variant, fine-tuned via reinforcement learning from human feedback (RLHF) for helpfulness and
 93 safety. The base model serves as the primary subject for direction extraction, following prior work.
 94 The instruct model allows us to examine how RLHF-induced alignment reshapes the geometry of the
 95 style direction. All experiments are run on a single NVIDIA A10G GPU via Modal cloud compute.

96 2.3 Direction Extraction

97 To extract the AI-style direction, we use two forms of Contrastive Activation Analysis: (1) Standard
 98 and (2) Detector-Weighted.

99 2.3.1 Standard CAA

100 For each text in the training split, we perform a forward pass and record the last-token residual stream
 101 activation $\mathbf{h}^{(l)} \in \mathbb{R}^{2048}$ at every layer $l \in \{0, \dots, 35\}$ where layer 0 is defined as the embedding
 102 layer. The CAA direction at layer l is the difference of class means:

$$\mathbf{d}_{\text{CAA}}^{(l)} = \frac{1}{N} \sum_{i=1}^N \mathbf{h}_{\text{AI},i}^{(l)} - \frac{1}{N} \sum_{i=1}^N \mathbf{h}_{\text{human},i}^{(l)} \quad (1)$$

103 where N is the number of pairs. We normalize each $\mathbf{d}_{\text{CAA}}^{(l)}$ to unit length and select the best layer l^* by
 104 maximizing linear classification accuracy on the validation set. The resulting unit vector $\hat{\mathbf{d}} = \frac{\mathbf{d}^{(l^*)}}{\|\mathbf{d}^{(l^*)}\|}$
 105 is used for all downstream ablation experiments unless otherwise specified.

106 2.3.2 Detector-Weighted CAA

107 Standard CAA treats all training examples equally, so the direction reflects an average contrast that
 108 may be influenced by non-stylistic distributional features. However, AI detectors implicitly encode
 109 a continuous notion of how "AI-sounding" a piece of text is, and a text that confidently fools an
 110 AI detector may carry less of the relevant stylistic signal than one flagged at 100% confidence. We
 111 therefore propose a detector-weighted variant of CAA:

$$\hat{\mathbf{d}}_{\text{weighted}}^{(l)} = \frac{\sum_i s_i \mathbf{h}_{\text{AI},i}^{(l)}}{\sum_i s_i} - \frac{\sum_i (1 - s_i) \mathbf{h}_{\text{human},i}^{(l)}}{\sum_i (1 - s_i)} \quad (2)$$

112 where $s_i \in [0, 1]$ is the detector’s AI-probability for the i -th training text. Detector-weighted CAA
 113 re-weights examples by the confidence of a pre-trained AI detector, so texts that the detector is more
 114 confident are AI-generated contribute more to the AI-side mean (have a higher weight), and texts the
 115 detector considers least AI-like contribute less. In other words, highly AI-like examples dominate the
 116 direction and ambiguous examples will contribute less.

117 We use three detectors with distinct decision boundaries for detector-weighted CAA. **(1)**
 118 **RoBERTa** [Liu et al., 2019] (openai-community/roberta-base-openai-detector) is a fine-
 119 tuned sequence classifier operating primarily on surface lexical distributional features (vocabulary
 120 register, punctuation patterns, n-gram statistics). **(2) RADAR** [Hu et al., 2023] is a DeBERTa-v3
 121 classifier trained in an adversarial loop against a Vicuna-7B paraphraser, making it robust to surface
 122 style-transfer and potentially sensitive to deeper syntactic and semantic patterns. **(3) GPTZero** [Tian
 123 and Cui, 2023] is a commercial detector combining per-sentence perplexity, burstiness, and a trained
 124 classifier, with its decision boundary most sensitive to local predictability.

125 2.4 Ablation Hook

126 To causally test the role of a layer or layer window, we register a persistent forward hook that
 127 orthogonally projects out the AI-style direction from the residual stream at every forward pass during
 128 generation:

$$\mathbf{h}_{\text{ablated}} = \mathbf{h} - (\mathbf{h} \cdot \hat{\mathbf{d}}) \hat{\mathbf{d}} \quad (3)$$

We evaluate three ablation conditions: (1) **Baseline**: no hook applied, (2) **Best-layer ablation**: hook at the single best-classification layer l^* only, (3) **All-layer ablation**: hook at all 36 layers simultaneously.

2.5 Layer-Window Sweep

For each window of $k \in \{1, 5\}$ consecutive layers with center position $c \in \{0, \dots, 35\}$, we apply the ablation hook only within that window and generate outputs. Sliding c across all 36 positions yields a causal profile of the style direction across the full network depth. The two choices of k correspond to qualitatively different interventions: $k = 1$ provides maximal localization, revealing fine-grained causal peaks, while $k = 5$ offers robustness against imprecise layer targeting and represents a larger intervention size. For each window position, we collect HC3 RoBERTa AI scores on 25 held-out HC3 prompt responses and use LLM-as-judge to obtain AI-likeness and writing-quality scores on responses to 20 writing prompts.

2.6 Evaluation

AI detection on HC3 test set. We report RoBERTa AI-probability scores $[0, 1]$ on held-out HC3 test responses. Human and AI reference scores on the HC3 test set are 0.036 and 0.870, respectively. For detector-weighted runs, we additionally report scores from the corresponding detector.

LLM-as-judge for writing prompts. We use Claude Sonnet 4.6 as an automated judge to rate outputs on AI-likeness (does this read as AI-generated?), writing quality (coherence, fluency, structure), and factual quality, each on a 1-7 scale. Ratings are collected on 20 MT-Bench-style writing prompts spanning five categories: creative writing, persuasive/argumentative, expository, narrative/reflective, and professional/functional. We report mean scores with standard deviations over $n = 20$ –25 samples per condition.

Capability benchmarks. We evaluate on four benchmarks: (1) MMLU (500 questions across 10 subjects, 5-shot), (2) HellaSwag (200 commonsense completion questions, 0-shot log-probability scoring), (3) TruthfulQA MC1 (200 truthfulness questions, 0-shot), and (4) WikiText-2 perplexity (200 held-out sentences).

Logit-lens vocabulary projection. We project the AI-style direction $\hat{\mathbf{d}}$ onto the model’s unembedding matrix $W_U \in \mathbb{R}^{V \times d}$ to produce a vocabulary-space signature $\mathbf{v} = W_U \hat{\mathbf{d}}$. Tokens with the largest positive entries in $\mathbf{v}$ are those whose generation probability would increase most if the direction were added to a residual stream hidden state; tokens with the most negative entries form the human-style vocabulary signature.

3 Results

3.1 The AI-style direction is linearly encoded and genuine

We first verify that a genuine AI-style direction exists in the residual stream independently of text length via standard CAA. Table 1 summarizes classification accuracy, AUC, and length cosine similarity across all four conditions: (original, length-matched HC3 datasets) $\times$ (QWEN2.5-3B, QWEN2.5-3B-INSTRUCT).

Table 1: Summary of extracted AI-style direction via standard CAA across four conditions. “Length cos. sim.” is the cosine similarity between the extracted direction and a pure text-length direction.

Condition	Best Layer	Test Acc. (%)	Test AUC	Length Cos. Sim.
Original (prior Hypogenic-AI work)	21	97.5	0.999	0.930
(1) QWEN2.5-3B (original HC3)	13	97.0	0.997	0.938
(2) QWEN2.5-3B-INSTRUCT (original HC3)	8	96.5	0.995	0.922
(3) QWEN2.5-3B (length-matched)	29	96.5	0.998	−0.065
(4) QWEN2.5-3B-INSTRUCT (length-matched)	29	98.0	0.998	−0.151

Length confound on original HC3. On the original HC3 dataset, the extracted CAA direction has cosine similarity 0.93 with a pure text-length direction, indicating that most of the classification

168 signal is attributable to verbosity rather than style: ChatGPT responses average 914 characters versus
169 446 for human responses. Figures 2–3 illustrate this confound. The projection distributions (Figure 2)
170 show clean separation between human and AI activations, but the scatter plot of text length versus
171 projection score (Figure 1) reveals that the separation tracks almost perfectly with response length.
172 The cosine similarity bar chart (Figure 2, right panel) confirms the 0.938 overlap with the length
173 direction.

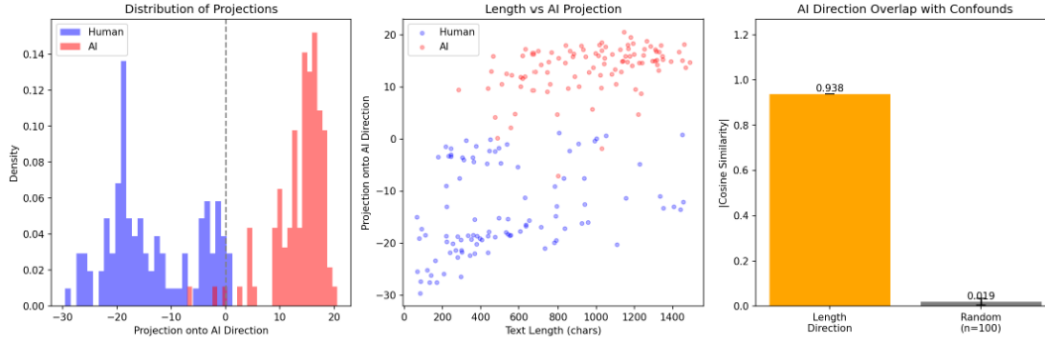


Figure 1: (1) QWEN2.5-3B (original HC3): (left) projection distributions onto the extracted CAA direction; (center) text length vs. projection score, revealing that the direction tracks response verbosity; (right) cosine similarity of the AI direction with the length direction (0.938) versus random directions.

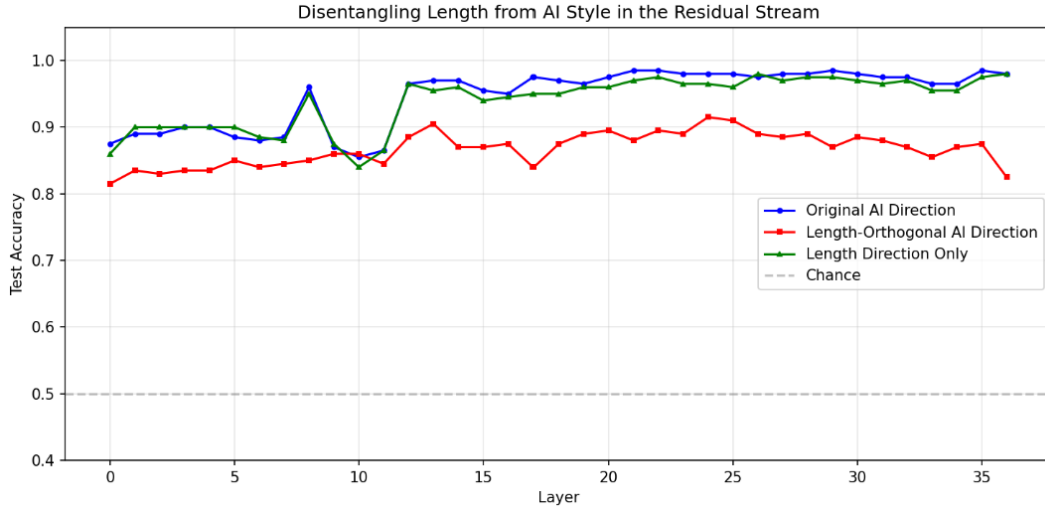


Figure 2: (1) QWEN2.5-3B (original HC3): classification accuracy by layer for the original AI direction, the length-orthogonal AI direction, and the length direction alone. The length direction alone achieves near-identical accuracy to the full AI Length, confirming verbosity as the dominant signal.

174 **Genuine style signal on length-matched HC3.** After length-matching, the cosine similarity between
175 the extracted direction and the length direction collapses to -0.065 for QWEN2.5-3B and -0.151
176 for QWEN2.5-3B-INSTRUCT which are near zero, confirming that verbosity has been removed as
177 a confounding signal. Critically, classification accuracy remains at 96.5% and 98.0% respectively,
178 demonstrating that a genuine stylistic signal independent of response length is linearly encoded in the
179 residual stream. Figures 5–6 show the corresponding analyses on the length-matched dataset. The
180 projection distributions remain well separated (Figure 5), the length-vs-projection scatter shows no
181 relationship (Figure 4), and PCA continues to reveal clean cluster structure across layers (Figure 6).

182 Two additional observations are worth noting. First, the best-classifying layer shifts from layer
183 13-21 on original HC3 to layer 29 on length-matched data for both models, suggesting that pure

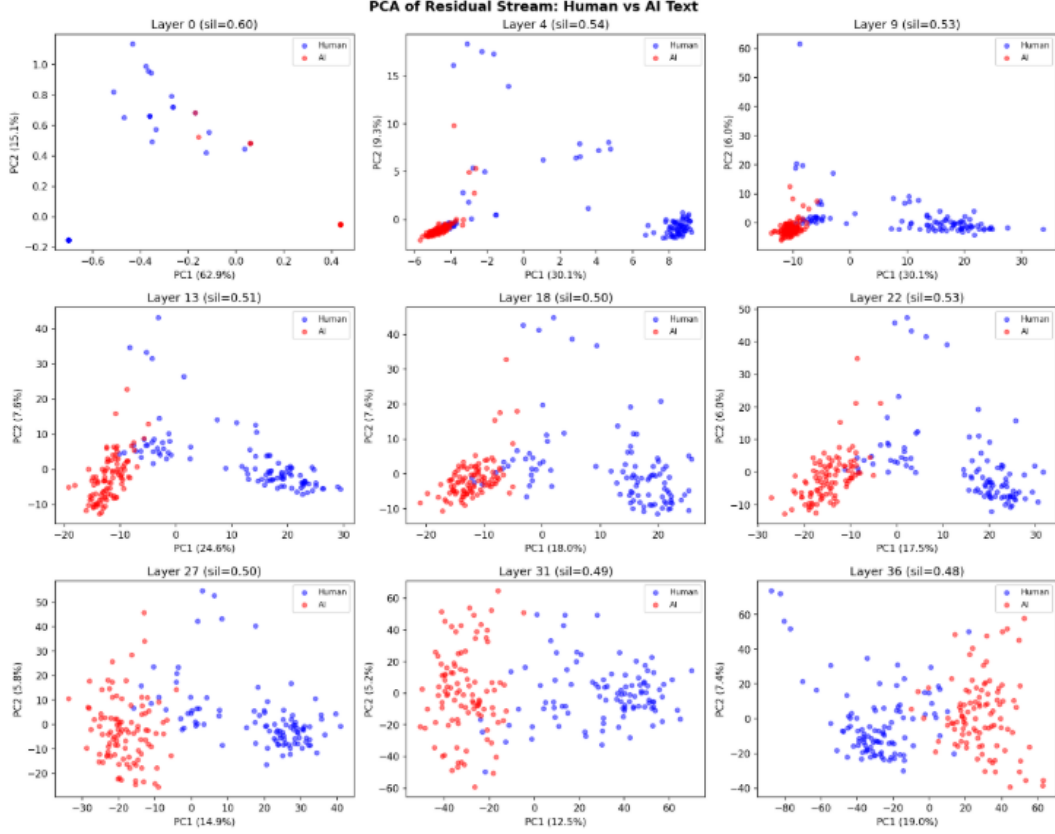


Figure 3: PCA of residual stream activations for (1) QWEN2.5-3B (original HC3) across selected layers. Human (blue) and AI (red) clusters separate clearly, but this separation is substantially driven by the length confound.

184 stylistic information is encoded deeper in the network once the early, surface-level length signal is
 185 removed. Second, QWEN2.5-3B-INSTRUCT achieves slightly higher accuracy than QWEN2.5-3B
 186 on length-matched data (98.0% vs. 96.5%), consistent with RLHF strengthening the stylistic signal
 187 independently of verbosity. All subsequent experiments in this paper use the length-matched HC3
 188 dataset unless otherwise specified.

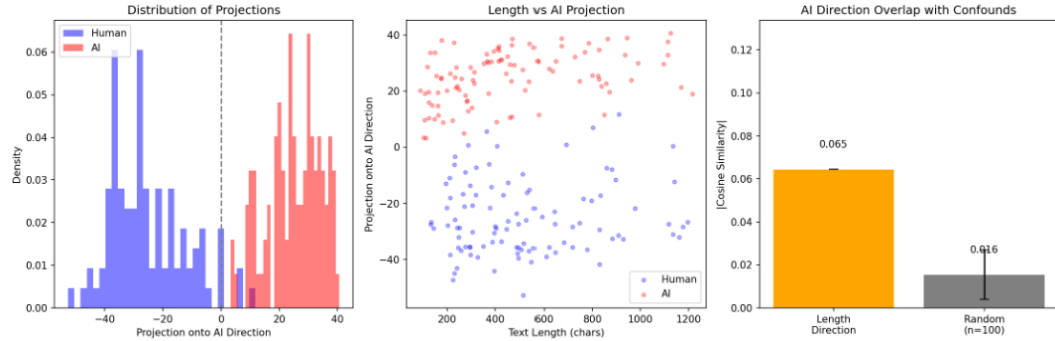


Figure 4: (3) QWEN2.5-3B (length-matched): (left) projection distributions onto the extracted CAA direction; (center) text length vs. projection score, showing no relationship between length and the extracted direction; (right) cosine similarity with the length direction collapses to 0.065, near that of random directions.

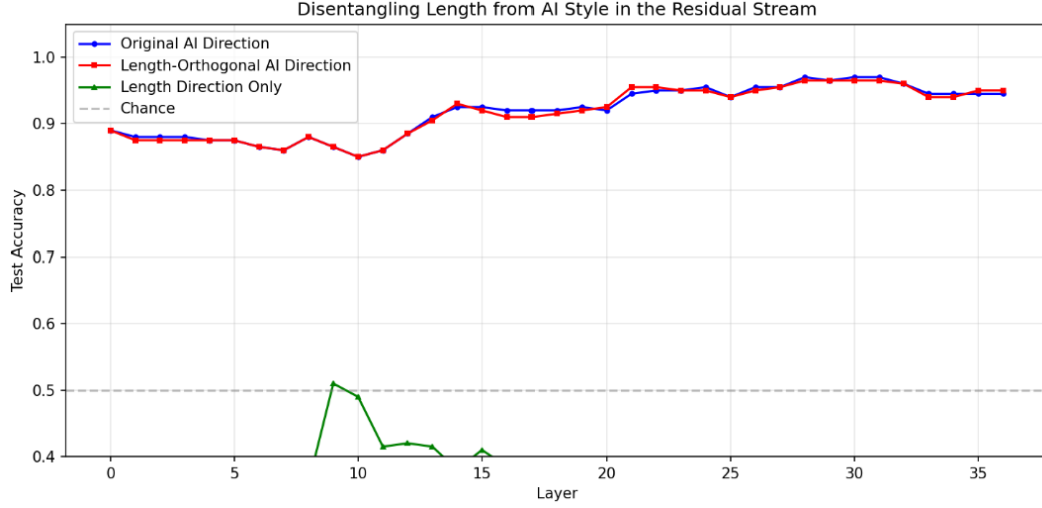


Figure 5: (3) QWEN2.5-3B (length-matched): classification accuracy by layer for the original AI direction and the length-orthogonal AI direction ((which now nearly coincide), alongside the length-only direction, which collapses to below-chance accuracy, confirming length carries no discriminative signal after matching.

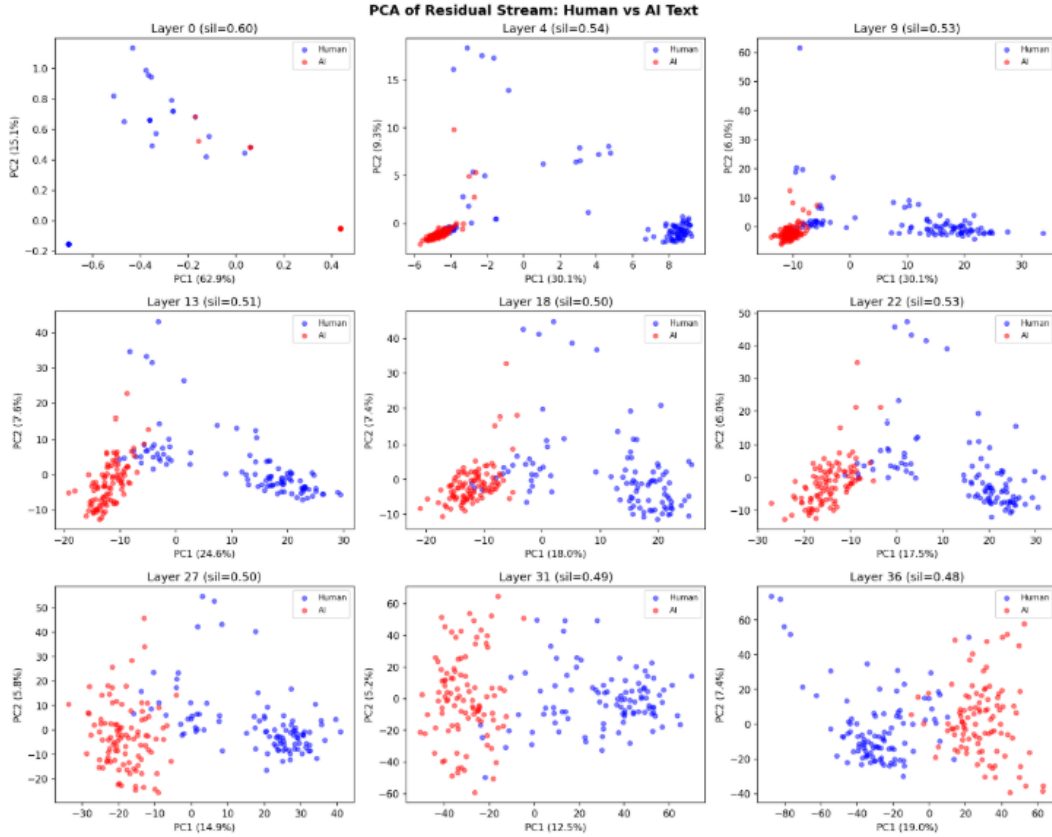


Figure 6: PCA of residual stream activations for (3) QWEN2.5-3B (length-matched) across selected layers. Human and AI clusters remain well separated, confirming that the linear structure reflects genuine stylistic differences rather than verbosity.

3.2 Representation-causation dissociation through layer sweep

Motivating observation. Initial ablation experiments revealed a striking null result: ablating the best-classifying layer ($l^* = 29$) during generation produces near-zero change in measured RoBERTa AI-likeness. Table 2 shows RoBERTa HC3 AI scores across initial ablation conditions for both models. Single-layer ablation at the best-classifying layer leaves the AI score essentially unchanged relative to baseline with a mean RoBERTa score of 0.196 for QWEN2.5-3B-INSTRUCT, approaching the human reference of 0.036. This dissociation between the layer that most linearly *encodes* the direction and the layers that causally *generate* AI style motivates the systematic sweep below.

Table 2: RoBERTa HC3 AI-probability scores (mean over 100 test outputs) for standard CAA on both models. Human reference = 0.036; AI reference = 0.870. Single-layer ablation at the best-classifying layer ($l^* = 29$) has essentially no effect, while all-layer ablation produces a substantial reduction.

Model	Baseline	Best-layer ablation	All-layer ablation
QWEN2.5-3B	0.840	0.871 (+0.031)	0.651 (−0.189)
QWEN2.5-3B-INSTRUCT	0.656	0.662 (+0.006)	0.196 (−0.460)

Layer sweep results. To identify where in the network AI style is causally generated, we conduct a systematic sliding-window sweep: for each window center $c \in \{0, \dots, 35\}$ and window size $k \in \{1, 5\}$, we apply the ablation hook only within that window and measure both RoBERTa HC3 AI score and LLM-judge AI-likeness and writing quality on 20 writing prompts. Figure 7 shows the full sweep across all four conditions: (QWEN2.5-3B, QWEN2.5-3B-INSTRUCT) $\times$ ($k = 1, k = 5$).

The results reveal three key patterns. First, ablating the best-classifying layer (28-29) produces near-zero change in AI score or LLM-judge AI-likeness across all conditions, confirming that the representation readout layer is causally inert. Second, a causal generation region consistently emerges in middle layers: for the base model, ablating around layer 15-19 ($k = 1$) reduces the RoBERTa HC3 AI score by approximately 37% relative to baseline, while for the instruct model the causal peak in HC3 score shifts slightly later. Third, for the writing-quality metric, the causal region is more consistently centered around layers 17-22 for both models, suggesting a partial dissociation between which layers drive HC3-style generation and which drive writing-register style.

Interpretation. One explanation for the seen representation-causation dissociation is that the final layers are a downstream reflector of style already committed by middle layers. Linear probing identifies where information is *encoded* most cleanly in the residual stream; it does not identify where that information *causally drives* behavior during generation. This distinction matters for mechanistic interpretability more broadly: high probing accuracy at a given layer is not sufficient evidence that the layer is a causal locus of the probed property.

3.3 Style suppression and the writing coherency tradeoff

While factual and commonsense benchmarks provide one lens on capability preservation, writing coherence is arguably the most directly relevant capability for AI-style interventions: the practical value of suppressing AI-likeness depends entirely on whether the resulting outputs remain readable and useful. A model that evades detection by producing incoherent text has not achieved a meaningful reduction in AI style. We therefore use writing coherence as the primary capability metric for characterizing the style-quality tradeoff, evaluated via LLM-as-judge scores on 20 MT-Bench-style writing prompts.

Figure 8 plots LLM-judge AI-likeness and writing quality on the same 1–7 axis for all four conditions (QWEN2.5-3B, QWEN2.5-3B-INSTRUCT) $\times$ ($k = 1, k = 5$), with a yellow band marking the most optimal style-quality tradeoff window. We informally define "sweet spot" to be the window where AI-likeness is most suppressed while coherence remains substantially above the all-layer ablation floor.

Coherence degradation under all-layer ablation. All-layer ablation achieves the deepest AI-likeness suppression (reaching 2.05-2.70 on the 7-point scale), but at a coherence cost that makes many outputs practically unusable. Qualitative inspection reveals three recurring failure modes: repetitive sentence structures (e.g., "*I'm not a baby. I'm not a child. I'm not a teenager...*"), structural fragmentation into numbered multiple-choice or bulleted list formats instead of prose, and

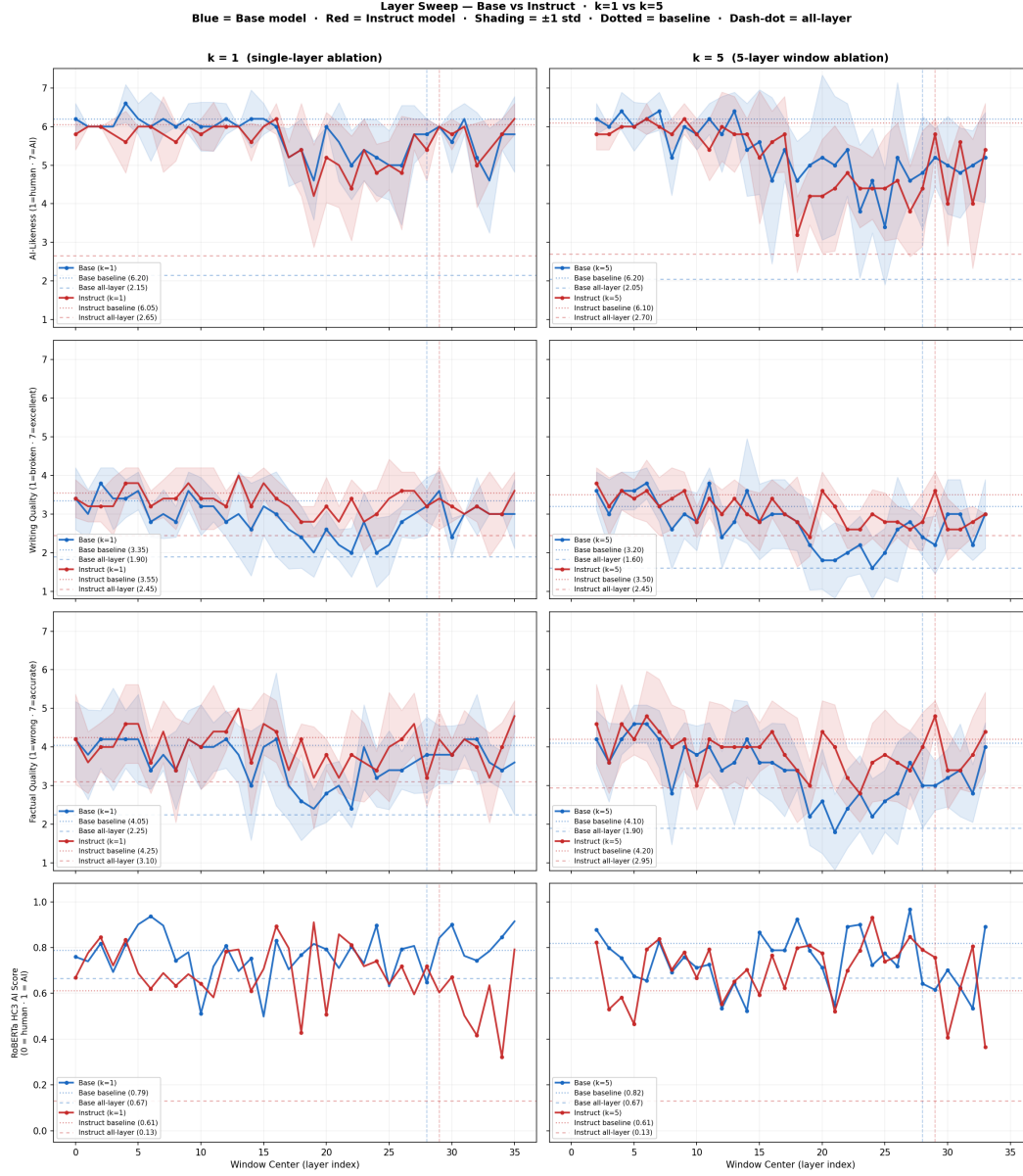


Figure 7: Layer sweep ablation results across all 36 layers for QWEN2.5-3B (blue) and QWEN2.5-3B-INSTRUCT (red), with $k = 1$ (left column) and $k = 5$ (right column). Each row shows a different evaluation metric: LLM-judge AI-likeness (top), writing quality (second), factual quality (third), and RoBERTa HC3 AI score (bottom). Dotted horizontal lines mark the unablated baseline; dash-dot lines mark all-layer ablation. Shaded regions are ± 1 SD. The best-classifying layer (28-29, gray dashed vertical line) consistently produces near-zero change in all metrics, while the causal generation region at middle layers 15-22 produces the largest reductions in AI-likeness.

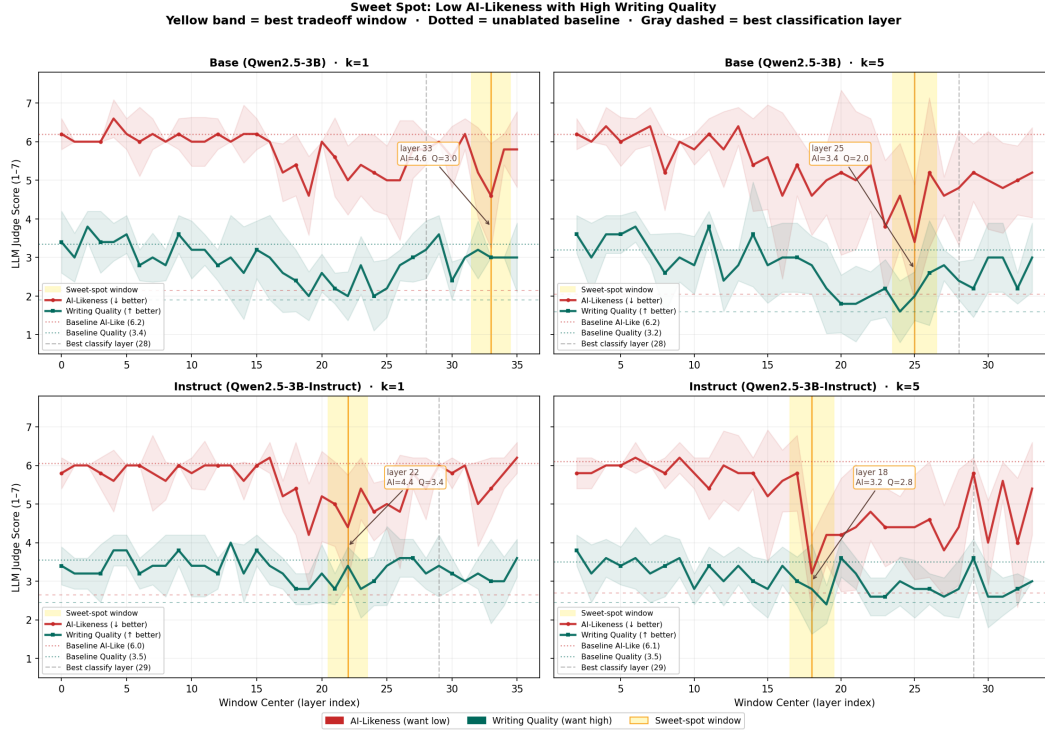


Figure 8: Sweet-spot visualization for all four conditions (QWEN2.5-3B and QWEN2.5-3B-INSTRUCT, $k = 1$ and $k = 5$). Red curve: LLM-judge AI-likeness (lower is better); green curve: writing quality (higher is better). Yellow band marks the sweet-spot window defined by the region where AI-likeness is most suppressed while writing quality is best preserved relative to all-layer ablation. Dotted lines mark the unablated baseline; gray dashed vertical line marks the best-classifying layer (28-29).

collapse into informal internet-register text (e.g., “*I think it happens because u have a lot of info in ur brain nd sometimes your brain gets confused*”). The base model’s degradation tends toward NLP benchmark artifacts, while the instruct model reverts toward conversational Reddit-style text, reflecting their different pretraining data distributions.

Coherence costs are not uniform across writing categories. Creative and narrative prompts preserve coherence better post-ablation, while professional and functional writing (resumes, cover letters, formal emails) degrades more heavily. This suggests that functional writing formats are more tightly coupled to the AI-style direction: their reliance on structured formatting conventions means that projecting out the style direction also disrupts the generation pathways responsible for producing that structure.

QWEN2.5-3B versus QWEN2.5-3B-INSTRUCT model tradeoff. Across both $k = 1$ and $k = 5$ conditions, QWEN2.5-3B-INSTRUCT consistently achieves a better style-writing coherence tradeoff than QWEN2.5-3B at the sweet spot window. For $k = 5$, the sweet spot for QWEN2.5-3B sits around layers 18-25, where LLM-judge AI-likeness drops from a baseline of 6.2 to approximately 3.4; writing quality, though reduced from baseline, is meaningfully higher than under all-layer ablation. QWEN2.5-3B-INSTRUCT reaches a similar AI-likeness reduction (3.2) at a comparable window centered around layer 18, but with better coherence preservation (writing quality = 2.8 vs. the base model’s ≈ 2.0). The gap between the two models suggests that the instruct model’s style representations are more compressible in the middle layers as projecting out the direction removes more of the stylistic signal while disturbing fewer of the generation pathways responsible for coherent output. This is consistent with RLHF producing a more factorized residual stream in which style and coherence are more orthogonally encoded, so that a targeted intervention on one does not strongly perturb the other.

Surgical window ablation as a practical middle ground. Surgical $k = 5$ window ablation substantially recovers coherence relative to all-layer ablation while maintaining AI-likeness reductions of ~ 3 points, making it the most practical intervention when output quality must be preserved. Single-layer ablation at the causal layer (e.g., layer 19 for QWEN2.5-3B) achieves more modest reductions with better coherence still, and may be preferable in settings where any coherence degradation is unacceptable. All-layer ablation remains useful as an upper bound on style suppression, but the coherence failures it induces suggest it is better understood as a diagnostic tool than a deployable intervention.

3.4 Detector-Weighted CAA: Ablation effectiveness, the style-writing coherency tradeoff, & MMLU, HellaSwag, and TruthfulQA preservation

Standard CAA weights all training examples equally, so the extracted direction reflects an average contrast that may be influenced by non-stylistic distributional features. Detector-weighted CAA re-weights by detector confidence so that prototypically AI-like examples dominate the direction. In this section, we analyze three aspects of detector-weighted CAA: (1) ablation effectiveness across detectors with distinct decision boundaries, (2) the style-writing coherence tradeoff under ablation, and (3) preservation of factual, commonsense, and truthfulness capabilities.

The best classification layer is largely consistent across detectors. Table 3 reports the best classification layer and HC3 AI scores across all eight runs (standard CAA and three DW-CAA variants) $\times$ (QWEN2.5-3B, QWEN2.5-3B-INSTRUCT). Across all detector-weighted runs 3-8, the detector used to extract the direction is used to score responses. Best classification layers are strikingly consistent across all three detectors, falling between layers 27-29 (75-81% of network depth) regardless of the weighting signal. This convergence suggests that all three detectors are responding to the same underlying representational structure in the residual stream despite their different characteristics.

Table 3: HC3 AI-probability scores. ($\pm\Delta$) denotes change from baseline.

Run	Detector	Best Layer	Baseline	Best-layer Ablation	All-layer Ablation	Human Ref.
(1) CAA Base	RoBERTa	29	0.840	0.871 (+0.031)	0.651 (−0.189)	0.036
(2) CAA Instruct	RoBERTa	29	0.656	0.662 (+0.006)	0.196 (−0.460)	0.036
(3) RoBERTa Base	RoBERTa	29	0.843	0.833 (−0.010)	0.612 (−0.231)	0.036
(4) RoBERTa Instruct	RoBERTa	29	0.716	0.707 (−0.009)	0.132 (−0.584)	0.036
(5) GPTZero Base	GPTZero	28	0.687	0.480 (−0.207)	0.089 (−0.598)	0.005
(6) GPTZero Instruct	GPTZero	28	0.910	0.756 (−0.154)	0.093 (−0.817)	0.005
(7) RADAR Base	RADAR	29	0.989	0.963 (−0.026)	0.952 (−0.037)	0.677
(8) RADAR Instruct	RADAR	27	0.946	0.900 (−0.046)	0.872 (−0.074)	0.677

Ablation effectiveness across detectors. Out of the three detectors, GPTZero seems to produce the most effective ablation as both GPTZero runs show dramatic score drops under all-layer ablation (base: -0.598 ; instruct: -0.817 delta). This suggests that GPTZero’s perplexity-based signals are particularly well-aligned with the generative causal subspace of the residual stream. On the other hand, RADAR is the least effective as scores remain very high across all conditions (-0.037 to -0.074 delta), likely reflecting a domain mismatch as RADAR was trained adversarially against Vicuna-7B paraphrases and may exploit features that do not correspond to those driving Qwen2.5-3B’s stylistic generation. Single-layer ablation at the best-classifying layer had minimal effect across all six DW-CAA runs, replicating the representation-causation dissociation from Section 3.2 and confirming it is not just an artifact of the standard CAA direction.

Style-writing coherence tradeoff across detectors. Across all eight writing evaluation runs, all-layer ablation brings AI-likeness to the 2.15-2.90 range from baselines of 6.10-6.25, which is a consistent reduction invariant of detector. As for writing coherency, base models consistently show larger writing coherence drops under all-layer ablation (coherence 1.60-2.00) while instruct models maintain substantially higher writing coherence (2.45-2.90) across all detector conditions, which was observed for standard CAA as well. This QWEN2.5-3B-INSTRUCT-QWEN2.5-3B-INSTRUCT

gap holding across both standard and detector-weighted CAA provides converging evidence that instruction fine-tuning geometrically separates the AI-style direction from coherence-preserving generation pathways.

With QWEN2.5-3B-INSTRUCT, RADAR achieves the best style-writing coherence tradeoff under all-layer ablation with writing coherence at 2.45 while AI-likeness drops to 2.65. One explanation is that RADAR’s adversarially-trained direction captures features that are more compressible in the instruct model’s residual stream and projecting it out removes style without strongly disrupting the generation pathways responsible for coherent writing structure. The base model shows no such RADAR advantage, consistent with the interpretation that the compressibility benefit is specific to the more factorized geometry produced by RLHF.

MMLU, HellaSwag, and TruthfulQA benchmarks. Figure 9 shows MMLU, HellaSwag, TruthfulQA, and WikiText-2 perplexity for the GPTZero DW-CAA run on QWEN2.5-3B-INSTRUCT, which is representative of the broader pattern across all detector conditions. MMLU and HellaSwag scores remain close to baseline under both best-layer and all-layer ablation across all runs, providing evidence that factual knowledge and commonsense reasoning are not co-located with the AI-style direction in at minimal the residual stream or are recoverable via other pathways when the direction is suppressed.

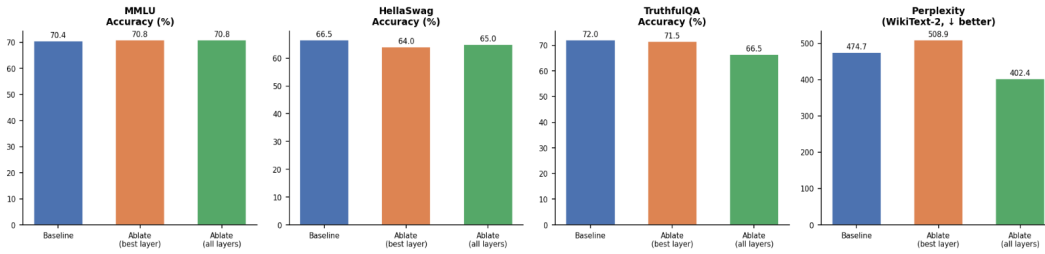


Figure 9: Capability benchmarks for GPTZero DW-CAA on QWEN2.5-3B-INSTRUCT under baseline, best-layer ablation, and all-layer ablation. MMLU and HellaSwag remain near baseline, while TruthfulQA stays stable for the instruct model. Perplexity decreases under all-layer ablation for the instruct model, indicating the style direction points partly away from the natural language distribution.

TruthfulQA reveals a slight base-instruct divergence. QWEN2.5-3B-INSTRUCT accuracy stays near baseline under all ablation conditions, consistent with RLHF explicitly training for truthfulness as a representation largely orthogonal to the style direction. The base model shows greater TruthfulQA degradation under all-layer ablation, suggesting that without explicit truthfulness training, truthful answering and AI-style occupy partially overlapping subspace in the residual stream, so removing the style direction partially disrupts truthful generation as a side effect. This degradation in TruthfulQA accuracy for QWEN2.5-3B is prominent in the case of RADAR and RoBERTa, but not with GPTZero.

WikiText-2 perplexity under ablation heavily varies by model type and ablation condition. For best-layer ablation, perplexity increases slightly across all detector conditions for both models except for GPTZero DW-CAA on QWEN2.5-3B. Under all-layer ablation, the pattern splits by model: for QWEN2.5-3B-INSTRUCT, perplexity decreases across all three detector conditions, while for QWEN2.5-3B, the direction depends on the detector. GPTZero produces a perplexity decrease, while RoBERTa and RADAR produce a slight increase. The instruct model result is interpretable in terms of RLHF: instruction tuning pushes the model toward formal, structured outputs that diverge from casual conversational text, and the trained style direction points partly away from the WikiText-2 distribution. Ablating it moves the instruct model back toward natural language, reducing perplexity. The base model’s detector dependence suggests that GPTZero’s perplexity-based signals may be more directly aligned with the WikiText-2 distribution than RoBERTa’s or RADAR’s surface-level features, so the GPTZero-weighted direction points more strongly away from natural text and its removal produces a similar decrease.

3.5 Vocabulary Signatures via Logit Lens

To better understand what the extracted AI-style direction encodes, we project it onto the model’s unembedding matrix W_U to produce a vocabulary-space signature $\mathbf{v} = W_U \hat{\mathbf{d}}$. Tokens with the largest positive entries are those whose generation probability would increase most if the direction were added to a residual stream hidden state. Tokens with the most negative entries form the human-style vocabulary signature. Results are broadly consistent across detector variants and model types.

Three interpretable clusters emerge in the AI-style tokens. First, emojis and markdown formatting tokens are characteristic of structured AI-generated content, where visual emphasis and bullet-point formatting are common. Second, Japanese formal constructions (, , meaning roughly "is able to" and "can do" in polite register) likely reflect Qwen’s multilingual pretraining, in which formal Japanese text is distributed similarly to formal English AI outputs in the residual stream. Third, web and code artifacts (creativecommons, @,) suggest the direction also picks up on structured reference and attribution conventions common in AI-generated explanatory text.

The human-style tokens form a coherent contrasting cluster: almost entirely informal English consisting of profanity, hedging qualifiers (maybe, probably, apparently), and emphatic colloquialisms (utterly, totally). This is consistent with the HC3 human responses originating predominantly from the ELI5 subreddit, where casual, opinionated, and colloquial language is the norm. The direction thus accurately captures the stylistic divergence between informal human discourse and polished AI outputs at the vocabulary level.

We note a limitation of this analysis: the logit-lens projection reveals which tokens are *correlated* with the AI-style direction in representation space, but does not establish that these tokens are causally responsible for AI-likeness in generated outputs. The direction may encode stylistic register at a more abstract level than individual token identity, with the vocabulary signature reflecting downstream consequences rather than proximate causes.

4 Discussion

AI-style versus refusal: a structural contrast. The representation-causation dissociation identified here stands in direct contrast with prior work on refusal [Arditi et al., 2024], where a single direction at one location mediates the behavior as ablating it changes model behavior cleanly and with low capability cost. AI-style does not behave this way. No single layer removal suffices to suppress it. The most linearly readable layer is causally inert as a window of middle layers is required for meaningful intervention. We hypothesize this difference reflects the distinct origins of the two properties. Refusal is a behaviorally trained binary switch, explicitly optimized by RLHF at a specific point in the network. AI-style, by contrast, is an emergent composite of many correlated generation patterns (formality, hedging, structure, and comprehensiveness) built up gradually across layers during pretraining on next-token prediction. This distributed origin makes it harder to localize and harder to ablate cleanly, and may explain why it does not admit the same single-direction, single-layer causal structure that refusal does.

Implications for AI text detection. The finding that AI style is causally generated in middle layers (15-22), rather than at the final readout layer, has practical implications for detector design. Detectors that probe final-layer representations may be reading a clean linear summary of a signal that was generated and committed several layers earlier. This raises the possibility that detection at intermediate layers could be both more causally informative and more robust to surface-level style transfer, since it would target the generative source rather than its downstream reflection.

Limitations and future work. Several limitations constrain the scope of our conclusions. All experiments use a single model family (Qwen2.5-3B and its instruct variant), and cross-model generalization has not been tested. The causal layer profile and sweet-spot window may differ across architectures and scales. The HC3 dataset contains only ChatGPT-generated AI text, so the extracted direction may not generalize to outputs from other LLMs such as Claude or Gemini, which have different stylistic signatures. LLM-as-judge evaluations are conducted with a maximum of 25 samples per window, resulting in high variance. Conclusions about specific sweet-spot layer positions should be treated as approximate rather than precise. Finally, the logit-lens vocabulary projection is correlational rather than causal. The tokens identified as AI-style or human-style vocabulary correlates of the direction are not confirmed to be causally responsible for AI-likeness in generated outputs.

Future work should address these limitations through activation patching to more precisely localize the generative mechanism, cross-model experiments to test generalization, and sparse autoencoder decomposition of the style direction into interpretable features that may reveal the multi-dimensional substructure of AI-style that a single direction cannot fully capture.

5 Conclusion

We have shown that the "AI-sounding" property of LLM outputs is genuinely linearly encoded in the residual stream, independent of text length, but that this linear encoding is causally distributed in a way that creates a sharp dissociation between representation and generation. The layer that most cleanly encodes the direction is causally inert during generation, while a distinct region of middle layers 15-22 constitutes the true causal generation zone. Surgical ablation of this region reduces AI-likeness by approximately 3 points on a 7-point scale, though always at some tradeoff with writing coherence, a cost that is generally lower for the instruction-tuned model than the base model.

These findings speak to a broader challenge in mechanistic interpretability: the tools we use to understand model internals (linear probing, difference-in-means directions, classification accuracy by layer) measure where information is *readable*, not where it *acts*. As LLMs become more deeply embedded in consequential domains, from academic integrity to content moderation to synthetic media governance, the gap between these two questions becomes practically important. Interventions designed to suppress or detect AI-style behavior need to target the generative source, not its downstream reflection. More broadly, understanding how stylistic, behavioral, and factual properties are encoded and causally distributed across transformer layers is a prerequisite for developing reliable, principled methods for model steering, AI detection, and alignment.

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452 A Writing Quality by Category

453 Figures 10 and 11 show the per-category breakdown of ablation effects across all six detector-weighted
454 runs. Figure 10 plots the drop in AI-likeness and coherence (baseline minus all-layer ablation) for
455 each writing category, making the differential costs visible across detectors. Figure 11 shows absolute
456 AI-likeness and coherence scores under all-layer ablation as a heatmap, with rows corresponding to
457 runs and columns to writing categories.

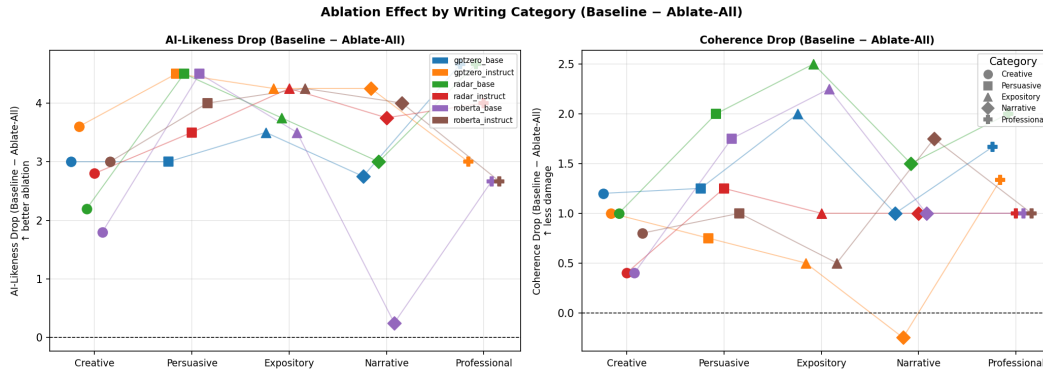


Figure 10: AI-likeness drop (left) and coherence drop (right) from baseline to all-layer ablation, broken down by writing category and detector run. Higher values indicate larger reductions. Creative and narrative categories show smaller coherence drops while persuasive and expository categories show larger AI-likeness reductions. Professional/functional writing is the most resistant to AI-likeness reduction, consistent with its structural formatting being tightly coupled to the style direction.

458 B Detector-Weighted CAA: Full LLM-Judge Results

459 Tables 4–7 report full LLM-judge scores (AI-likeness, coherence, factual) for all six detector-weighted
460 runs across baseline, best-layer ablation, and all-layer ablation conditions, for both HC3 and writing
461 prompt evaluations. Negative steering results are omitted as the focus of this work is on ablation.

462 C Layer Sweep: Full Window Results

463 Tables 8–11 report LLM-judge AI-likeness, coherence, and factual scores for every window position
464 in the layer sweep, for all four conditions (QWEN2.5-3B, QWEN2.5-3B-INSTRUCT) $\times$ ($k = 1$,
465 $k = 5$). Reference conditions (baseline and all-layer ablation) are included at the top of each table.
466 $n = 5$ per window position.

Table 4: LLM-judge scores on HC3 prompts — base models ($n = 25$ per condition). Values are mean $\pm$ std.

Run	Condition	AI-likeness	Coherence	Factual
3*RoBERTa Base	Baseline	5.52 $\pm$ 1.06	3.16 $\pm$ 1.59	3.00 $\pm$ 1.67
	Best-layer ablation	4.56 $\pm$ 1.30	2.44 $\pm$ 1.33	2.08 $\pm$ 1.29
	All-layer ablation	2.64 $\pm$ 1.60	1.88 $\pm$ 0.99	2.04 $\pm$ 1.04
3*RADAR Base	Baseline	5.24 $\pm$ 1.14	3.20 $\pm$ 1.30	2.92 $\pm$ 1.44
	Best-layer ablation	4.68 $\pm$ 1.32	2.40 $\pm$ 1.33	2.20 $\pm$ 1.20
	All-layer ablation	2.60 $\pm$ 1.67	1.32 $\pm$ 0.55	1.60 $\pm$ 1.17
3*GPTZero Base	Baseline	5.24 $\pm$ 1.34	2.88 $\pm$ 1.51	2.84 $\pm$ 1.62
	Best-layer ablation	3.96 $\pm$ 1.68	2.20 $\pm$ 0.80	2.32 $\pm$ 1.22
	All-layer ablation	2.28 $\pm$ 1.31	1.32 $\pm$ 0.61	1.36 $\pm$ 0.74

Table 5: LLM-judge scores on HC3 prompts — instruct models ($n = 25$ per condition). Values are mean $\pm$ std.

Run	Condition	AI-likeness	Coherence	Factual
3*RoBERTa Instruct	Baseline	5.48 $\pm$ 0.98	3.36 $\pm$ 0.97	3.00 $\pm$ 1.13
	Best-layer ablation	4.76 $\pm$ 1.14	2.88 $\pm$ 1.11	2.64 $\pm$ 1.09
	All-layer ablation	2.48 $\pm$ 1.20	2.24 $\pm$ 0.86	2.12 $\pm$ 1.21
3*RADAR Instruct	Baseline	5.80 $\pm$ 0.89	3.04 $\pm$ 0.82	2.64 $\pm$ 1.09
	Best-layer ablation	5.28 $\pm$ 1.00	2.80 $\pm$ 0.75	2.68 $\pm$ 1.26
	All-layer ablation	2.56 $\pm$ 0.85	2.60 $\pm$ 0.75	2.36 $\pm$ 0.84
3*GPTZero Instruct	Baseline	5.68 $\pm$ 0.93	3.48 $\pm$ 1.06	3.68 $\pm$ 1.52
	Best-layer ablation	4.72 $\pm$ 1.25	2.72 $\pm$ 0.87	2.80 $\pm$ 0.98
	All-layer ablation	2.16 $\pm$ 0.73	2.08 $\pm$ 0.56	2.20 $\pm$ 0.85

Table 6: LLM-judge scores on writing prompts — base models ($n = 20$ per condition). Values are mean $\pm$ std.

Run	Condition	AI-likeness	Coherence	Factual
3*RoBERTa Base	Baseline	6.15 $\pm$ 0.57	3.25 $\pm$ 0.83	4.10 $\pm$ 0.99
	Best-layer ablation	6.05 $\pm$ 0.59	3.20 $\pm$ 0.68	3.80 $\pm$ 0.98
	All-layer ablation	3.65 $\pm$ 1.74	2.00 $\pm$ 0.95	2.60 $\pm$ 1.16
3*RADAR Base	Baseline	6.20 $\pm$ 0.40	3.40 $\pm$ 0.58	4.10 $\pm$ 0.77
	Best-layer ablation	6.00 $\pm$ 0.71	3.20 $\pm$ 0.75	3.75 $\pm$ 1.04
	All-layer ablation	2.70 $\pm$ 1.73	1.65 $\pm$ 0.73	2.00 $\pm$ 0.77
3*GPTZero Base	Baseline	6.15 $\pm$ 0.48	3.20 $\pm$ 0.60	4.10 $\pm$ 0.77
	Best-layer ablation	5.85 $\pm$ 0.65	2.80 $\pm$ 0.87	3.55 $\pm$ 1.16
	All-layer ablation	2.85 $\pm$ 1.74	1.80 $\pm$ 0.87	2.10 $\pm$ 0.99

Table 7: LLM-judge scores on writing prompts — instruct models ($n = 20$ per condition). Values are mean $\pm$ std.

Run	Condition	AI-likeness	Coherence	Factual
3*RoBERTa Instruct	Baseline	6.10 $\pm$ 0.44	3.65 $\pm$ 0.48	4.45 $\pm$ 0.80
	Best-layer ablation	5.90 $\pm$ 0.54	3.50 $\pm$ 0.81	4.15 $\pm$ 0.85
	All-layer ablation	2.50 $\pm$ 1.24	2.65 $\pm$ 0.65	3.30 $\pm$ 0.90
3*RADAR Instruct	Baseline	6.25 $\pm$ 0.43	3.35 $\pm$ 0.57	4.25 $\pm$ 0.83
	Best-layer ablation	6.10 $\pm$ 0.54	3.40 $\pm$ 0.58	4.10 $\pm$ 0.70
	All-layer ablation	2.65 $\pm$ 1.24	2.45 $\pm$ 0.80	2.95 $\pm$ 1.02
3*GPTZero Instruct	Baseline	6.10 $\pm$ 0.44	3.50 $\pm$ 0.81	4.40 $\pm$ 0.86
	Best-layer ablation	5.65 $\pm$ 0.73	3.15 $\pm$ 0.65	3.90 $\pm$ 0.99
	All-layer ablation	2.15 $\pm$ 0.85	2.85 $\pm$ 0.36	3.35 $\pm$ 0.73

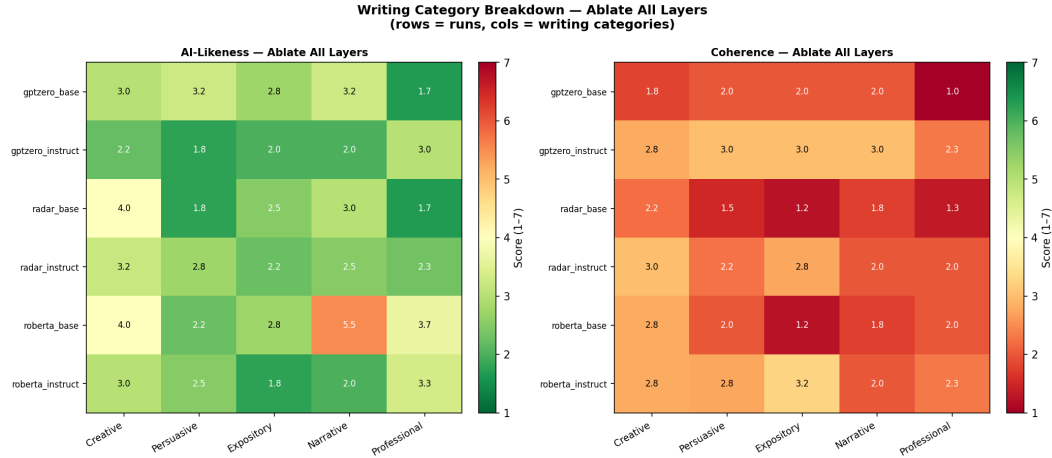


Figure 11: Writing category breakdown under all-layer ablation across all six detector-weighted runs (rows) and five writing categories (columns). Left: AI-likeness scores (green = low = more human-like); right: coherence scores (green = high = better preserved). Professional/functional writing retains the highest AI-likeness scores post-ablation, while expository and persuasive categories achieve the deepest AI-likeness reductions. Coherence is most preserved in narrative and creative categories.

D Ablated Writing Samples

The following examples illustrate qualitative output changes under all-layer ablation for the GPTZero QWEN2.5-3B-INSTRUCT run, selected to represent three characteristic patterns: coherence collapse in creative writing, degradation in explanatory writing, and resistance to ablation in professional formats.

Creative writing: poem about rain. Prompt: “Write a short poem about the feeling of rain on a summer evening.”

Baseline (GPTZero score: 1.000):

The gentle rain starts to fall, / On a summer evening’s calm. / It brushes against my skin, / A cool, refreshing feel. / The air grows fresh and clean, / As droplets dance in light. / I close my eyes and breathe it in, / This moment, so divine. / The world around me seems to slow, / As nature’s symphony plays. / And I am lost in its embrace, / In this quiet, peaceful haze.

All-layer ablation (GPTZero score: 0.003):

Rain on the summery eve / Drops of purest, softest glee / Lullaby of a summer night / I’ll not be going out tonight / The cool air in my nostrils fills / Bursts of life through every cell / A world so very nice / [...] Oh, Summer! I can’t wait till next time / To feel your soft, wet kiss on me / A new day just begins / And you are here to stay [output trails off mid-thought]

The ablated output shifts from structured, balanced stanzas toward fragmented, more colloquial phrasing. The GPTZero score drops from 1.000 to 0.003, reflecting a near-complete suppression of the perplexity-based AI signal, but the output also loses the formal prosodic structure of the baseline.

Explanatory writing: concept of time. Prompt: “Explain the concept of time to someone who has never encountered the idea.”

Baseline (GPTZero score: 1.000):

Time is a measure of duration or progression, often measured in units such as seconds, minutes, hours, days, weeks, months, and years. It is a fundamental concept that helps us organize our lives and understand the passage of events. [...] Time can also be thought of as a flow, like water flowing through a stream.

Table 8: Layer sweep results: QWEN2.5-3B, $k = 5$. Best classification layer: 28. Baseline: AI=6.20, Coh=3.20, Fact=4.10. All-layer: AI=2.05, Coh=1.60, Fact=1.90.

Window	Center	AI-likeness	Coherence	Factual
0	2	6.20±0.40	3.60±0.49	4.20±0.75
1	3	6.00±0.00	3.00±0.89	3.60±1.02
2	4	6.40±0.49	3.60±0.49	4.20±0.40
3	5	6.00±0.63	3.60±0.49	4.60±0.49
4	6	6.20±0.40	3.80±0.40	4.60±0.49
5	7	6.40±0.49	3.20±0.40	4.20±0.75
6	8	5.20±0.98	2.60±1.02	2.80±1.33
7	9	6.00±0.00	3.00±0.63	4.00±0.63
8	10	5.80±0.40	2.80±0.75	3.80±0.75
9	11	6.20±0.40	3.80±0.40	4.00±0.63
10	12	5.80±0.98	2.40±0.80	3.40±0.49
11	13	6.40±0.49	2.80±0.40	3.60±1.62
12	14	5.40±1.20	3.60±1.36	4.20±0.98
13	15	5.60±1.36	2.80±0.98	3.60±0.80
14	16	4.60±2.15	3.00±0.89	3.60±1.02
15	17	5.40±0.80	3.00±0.89	3.40±1.20
16	18	4.60±1.02	2.80±0.75	3.40±0.49
17	19	5.00±0.89	2.20±1.17	2.20±0.75
18	20	5.20±2.13	1.80±0.75	2.60±1.36
19	21	5.00±1.79	1.80±0.98	1.80±0.98
20	22	5.40±1.20	2.00±0.63	2.40±1.02
21	23	3.80±1.72	2.20±0.75	2.80±0.98
22	24	4.60±1.36	1.60±0.80	2.20±0.98
23	25	3.40±1.50	2.00±0.63	2.60±0.80
24	26	5.20±1.94	2.60±1.36	2.80±1.33
25	27	4.60±0.49	2.80±0.40	3.60±0.80
26	28	4.80±1.47	2.40±0.49	3.00±1.09
27	29	5.20±0.75	2.20±0.75	3.00±0.63
28	30	5.00±1.26	3.00±0.89	3.20±0.75
29	31	4.80±1.17	3.00±0.89	3.40±1.20
30	32	5.00±0.89	2.20±0.40	2.80±0.75
31	33	5.20±1.17	3.00±0.89	4.00±0.63

496 *Just as the water moves from one place to another, so does time move from one*
497 *moment to the next.*

498 **All-layer ablation** (GPTZero score: 0.002):

499 *Time is a thing that can be measured and it goes forward, and can not be reversed.*
500 *[...] I am 31 years old means I have been alive for 31 years [...] You are always*
501 *10 minutes younger than you were last week. You are a 12 year old boy. Imagine*
502 *you have a friend that has a bike, you ask him if he could lend you his bike and*
503 *ride with you to get ice cream, you will say something like “when are you free?”*

504 The ablated output becomes informal and concrete but retains topical relevance. The shift toward
505 colloquial examples (bike rides, ice cream) is consistent with the model reverting to Reddit/ELI5-style
506 text under ablation.

507 **Professional writing: job offer decline email.** Prompt: “Write a professional email declining a job
508 offer while maintaining a positive relationship.”

509 **Baseline** (GPTZero score: 0.999):

510 *Subject: Declining the Job Offer / Dear [Hiring Manager’s Name], / I hope this*
511 *message finds you well. I am writing to express my gratitude for the opportunity*
512 *to interview and consider a position with [Company Name]. [...] After careful*
513 *consideration of all aspects, including the details of the offer, my current situation,*
514 *and some recent personal developments, I have decided that it would be best for*
515 *me to decline the position at this time.*

Table 9: Layer sweep results: QWEN2.5-3B-INSTRUCT, $k = 5$. Best classification layer: 29. Baseline: AI=6.10, Coh=3.50, Fact=4.20. All-layer: AI=2.70, Coh=2.45, Fact=2.95.

Window	Center	AI-likeness	Coherence	Factual
0	2	5.80±0.40	3.80±0.40	4.60±1.02
1	3	5.80±0.40	3.20±0.75	3.60±0.80
2	4	6.00±0.00	3.60±0.49	4.60±1.02
3	5	6.00±0.00	3.40±0.49	4.20±0.40
4	6	6.20±0.40	3.60±0.80	4.80±1.17
5	7	6.00±0.00	3.20±0.75	4.40±1.36
6	8	5.80±0.40	3.40±0.80	4.00±1.09
7	9	6.20±0.40	3.60±0.49	4.20±0.75
8	10	5.80±0.40	2.80±0.40	3.00±0.63
9	11	5.40±0.80	3.40±0.49	4.20±0.98
10	12	6.00±0.89	3.00±0.63	4.00±1.09
11	13	5.80±0.98	3.40±0.49	4.00±0.89
12	14	5.80±0.40	3.00±0.63	4.00±1.09
13	15	5.20±1.72	2.80±0.75	4.00±0.63
14	16	5.60±0.80	3.40±0.49	4.40±0.80
15	17	5.80±0.98	3.00±0.63	3.80±1.17
16	18	3.20±0.98	2.80±1.17	3.40±1.36
17	19	4.20±0.75	2.40±0.49	3.00±0.63
18	20	4.20±1.47	3.60±0.49	4.40±0.80
19	21	4.40±1.62	3.20±0.40	4.00±1.09
20	22	4.80±0.75	2.60±0.49	3.20±0.75
21	23	4.40±1.02	2.60±0.49	2.80±0.75
22	24	4.40±0.80	3.00±0.63	3.60±1.02
23	25	4.40±1.36	2.80±0.75	3.80±1.17
24	26	4.60±1.50	2.80±0.40	3.60±1.02
25	27	3.80±1.17	2.60±0.49	3.40±1.02
26	28	4.40±1.50	2.80±0.75	4.00±0.89
27	29	5.80±0.40	3.60±0.49	4.80±0.40
28	30	4.00±1.09	2.60±0.80	3.40±1.02
29	31	5.60±0.49	2.60±0.49	3.40±0.49
30	32	4.00±1.67	2.80±0.40	3.80±0.98
31	33	5.40±1.20	3.00±0.00	4.40±1.02

516 **All-layer ablation** (GPTZero score: 0.997):

517 *Subject: No thanks for the position / Hi [Hr name] / I have been thinking about*
518 *your job offer for quite some time, and I really appreciate the opportunity. After*
519 *much deliberation, I have decided to not take up this offer. [...] So please accept*
520 *my apologies for not accepting this offer but I hope you understand and that we*
521 *can stay as friends / colleagues etc.*

522 This example illustrates the professional/functional category’s resistance to ablation: despite all-layer
523 projection, the GPTZero score barely moves (0.999 → 0.997). The output does shift from formal to
524 casual register (“Hi [Hr name]”, “friends / colleagues etc.”), but the email structure itself persists,
525 consistent with the finding in Appendix A that professional formats are among the most tightly
526 coupled to the AI-style direction.

527 E LLM-as-Judge Prompts

528 HC3 / ELI5 Evaluation Prompt

529 You are evaluating responses to questions from the HC3 dataset (Reddit’s r/explainlikeimfive
530 subreddit). These are questions where people ask for accessible explanations of complex
531 topics. Responses can range from casual Reddit-style answers to formal AI-generated
532 explanations.
533 Original question: {prompt}
534 Generated response: {response}

Table 10: Layer sweep results: QWEN2.5-3B, $k = 1$. Best classification layer: 28. Baseline: AI=6.20, Coh=3.35, Fact=4.05. All-layer: AI=2.15, Coh=1.90, Fact=2.25.

Window	Center	AI-likeness	Coherence	Factual
0	0	6.20±0.40	3.40±0.80	4.20±0.98
1	1	6.00±0.00	3.00±0.63	3.80±1.17
2	2	6.00±0.00	3.80±0.40	4.20±0.75
3	3	6.00±0.00	3.40±0.80	4.20±1.33
4	4	6.60±0.49	3.40±0.49	4.20±0.75
5	5	6.20±0.40	3.60±0.49	4.20±1.17
6	6	6.00±0.89	2.80±0.40	3.40±0.49
7	7	6.20±0.40	3.00±0.63	3.80±0.40
8	8	6.00±0.00	2.80±0.40	3.40±1.02
9	9	6.20±0.40	3.60±0.49	4.20±0.75
10	10	6.00±0.63	3.20±0.75	4.00±1.09
11	11	6.00±0.63	3.20±0.40	4.00±0.63
12	12	6.20±0.40	2.80±0.40	4.20±0.75
13	13	6.00±0.00	3.00±1.09	3.80±0.75
14	14	6.20±0.75	2.60±1.20	3.00±1.41
15	15	6.20±0.40	3.20±0.75	4.00±0.63
16	16	6.00±0.00	3.00±1.09	4.20±1.72
17	17	5.20±0.75	2.60±1.02	3.00±0.89
18	18	5.40±0.80	2.40±0.80	2.60±1.02
19	19	4.60±1.02	2.00±0.63	2.40±0.49
20	20	6.00±0.00	2.60±0.49	2.80±0.75
21	21	5.60±1.02	2.20±0.40	3.00±0.63
22	22	5.00±0.89	2.00±0.63	2.40±0.49
23	23	5.40±0.80	2.80±0.75	4.00±1.09
24	24	5.20±0.98	2.00±0.89	3.20±0.75
25	25	5.00±0.89	2.20±0.75	3.40±1.02
26	26	5.00±1.55	2.80±0.40	3.40±0.49
27	27	5.80±0.75	3.00±0.63	3.60±0.80
28	28	5.80±0.40	3.20±0.75	3.80±0.98
29	29	6.00±0.00	3.60±0.49	3.80±0.75
30	30	5.60±0.80	2.40±0.49	3.80±0.75
31	31	6.20±0.40	3.00±0.00	4.20±0.40
32	32	5.20±1.17	3.20±0.75	4.20±1.17
33	33	4.60±1.36	3.00±0.00	3.60±0.49
34	34	5.80±0.40	3.00±0.00	3.40±0.49
35	35	5.80±0.98	3.00±0.89	3.60±1.36

Rate this response on three dimensions using the anchored scales below.

AI-LIKENESS (1-7):

1 = Clearly human: casual, informal, conversational, short, may have typos

2 = Mostly human: informal but structured

3 = Slightly human: some informal elements, some structure

4 = Mixed: formal and informal elements in roughly equal measure

5 = Slightly AI: mostly formal/structured with occasional casual phrasing

6 = Mostly AI: formal, comprehensive, well-organised, hedging language

7 = Clearly AI: highly formal, bullet-pointed, "I'd be happy to help", comprehensive coverage, textbook-style explanations

COHERENCE (1-7):

1 = Completely incoherent, off-topic, or clearly broken (e.g. repeating text, format collapse into multiple-choice questions)

4 = Mostly on-topic but with noticeable gaps or tangents

7 = Clear, on-topic, well-organised, directly answers the question

FACTUAL REASONABLENESS (1-7):

1 = Clearly wrong, nonsensical, or contains major factual errors

4 = Mostly correct with minor inaccuracies

7 = Accurate, informative, no obvious errors

Respond with ONLY a JSON object, no other text: {"ai_likeness": <1-7>, "coherence": <1-7>, "factual": <1-7>}

Table 11: Layer sweep results: QWEN2.5-3B-INSTRUCT, $k = 1$. Best classification layer: 29.
Baseline: AI=6.05, Coh=3.55, Fact=4.25. All-layer: AI=2.65, Coh=2.45, Fact=3.10.

Window	Center	AI-likeness	Coherence	Factual
0	0	5.80±0.40	3.40±0.49	4.20±1.17
1	1	6.00±0.00	3.20±0.40	3.60±0.80
2	2	6.00±0.00	3.20±0.40	4.00±0.63
3	3	5.80±0.40	3.20±0.75	4.00±0.89
4	4	5.60±0.80	3.80±0.40	4.60±1.02
5	5	6.00±0.00	3.80±0.40	4.60±1.02
6	6	6.00±0.00	3.20±0.40	3.60±0.80
7	7	5.80±0.98	3.40±0.49	4.40±0.80
8	8	5.60±0.49	3.40±0.80	3.40±1.36
9	9	6.00±0.00	3.80±0.40	4.20±0.40
10	10	5.80±0.40	3.40±0.80	4.00±1.09
11	11	6.00±0.63	3.40±0.80	4.40±1.20
12	12	6.00±0.00	3.20±0.40	4.40±0.49
13	13	6.00±0.00	4.00±0.00	5.00±0.00
14	14	5.60±0.49	3.20±0.75	3.60±1.36
15	15	6.00±0.00	3.80±0.40	4.60±0.80
16	16	6.20±0.40	3.40±0.49	4.40±0.80
17	17	5.20±0.40	3.20±0.40	3.40±0.49
18	18	5.40±1.20	2.80±0.40	4.20±0.40
19	19	4.20±1.33	2.80±0.40	3.20±1.33
20	20	5.20±1.17	3.20±0.75	3.80±0.40
21	21	5.00±1.09	2.80±0.40	3.20±0.40
22	22	4.40±1.36	3.40±0.49	3.80±1.17
23	23	5.40±0.80	2.80±0.75	3.60±1.02
24	24	4.80±0.75	3.00±0.63	3.40±0.49
25	25	5.00±0.63	3.40±1.02	4.00±1.41
26	26	4.80±1.47	3.60±0.49	4.20±0.75
27	27	5.80±0.40	3.60±0.49	4.60±0.80
28	28	5.40±0.80	3.20±0.40	3.20±0.75
29	29	6.00±0.00	3.40±0.80	4.20±0.75
30	30	5.80±0.40	3.20±0.40	3.80±0.40
31	31	6.00±0.00	3.00±0.00	4.20±0.75
32	32	5.00±1.09	3.20±0.40	4.00±0.63
33	33	5.40±0.80	3.00±1.09	3.20±1.17
34	34	5.80±0.40	3.00±0.63	4.00±0.63
35	35	6.20±0.40	3.60±0.49	4.80±0.40

Writing Prompt Evaluation

You are evaluating responses to creative and expository writing prompts.

Writing prompt: {prompt}

Generated response: {response}

AI-LIKENESS (1–7):

1 = Clearly human: casual, idiosyncratic, natural imperfections, personal voice

2 = Mostly human: personal voice with minor formal elements

3 = Slightly human: some natural phrasing but noticeable polish

4 = Mixed: some human touches, some AI-typical patterns in roughly equal measure

5 = Slightly AI: mostly polished/structured with occasional natural phrasing

6 = Mostly AI: uniformly polished, generic metaphors, hedging, over-completeness

7 = Clearly AI: highly formulaic structure, exhaustive coverage, “certainly” / “I’d be happy to”, perfectly balanced sentences, textbook-style prose

WRITING QUALITY (1–7):

1 = Completely broken: incoherent, off-topic, format collapse, repetition

2 = Very poor: mostly off-topic or barely readable

3 = Poor: several noticeable problems but some coherence

4 = Adequate: addresses the prompt but writing is flat or generic

5 = Good: addresses the prompt with reasonable style and clarity

6 = Very good: engaging, clear, shows craft in language choice

7 = Excellent: vivid, distinctive, well-crafted, effectively addresses the prompt

577 **FACTUAL / CREATIVE REASONABLENESS (1-7):**
578 For factual writing: 1 = clear errors, 7 = accurate and well-grounded
579 For creative writing: 1 = incoherent or contradictory, 7 = internally consistent, imaginative,
580 and effective
581 Respond with ONLY a JSON object, no other text: {"ai_likeness": <1-7>, "coherence":
582 <1-7>, "factual": <1-7>}

583 **F Writing Evaluation Prompts**

584 The following 20 MT-Bench-style prompts were used for all writing evaluations, spanning five
585 categories of four prompts each.

586 **Creative Writing.**

- 587 1. Write a short poem about the feeling of rain on a summer evening.
- 588 2. Write the opening paragraph of a mystery novel set in a lighthouse.
- 589 3. Write a short story about a child who discovers that their shadow has a mind of its own.
- 590 4. Create a myth explaining why the sky changes colors at sunset.
- 591 5. Write a product review for the concept of 'sleep' as if it were a new consumer technology.

592 **Persuasive / Argumentative.**

- 593 6. Write a persuasive argument for why everyone should learn to cook their own meals.
- 594 7. Write a speech encouraging high school students to pursue careers in science.
- 595 8. Write a brief manifesto for a society that values rest and slowness over productivity.
- 596 9. Write a letter to your city council arguing that more public green spaces should be built.

597 **Expository / Explanatory.**

- 598 10. Explain the concept of time to someone who has never encountered the idea.
- 599 11. Describe what makes a city feel alive, drawing on sensory and social details.
- 600 12. Explain how music can change a person's emotional state, using concrete examples.
- 601 13. Describe the water cycle as if you were telling a story to a curious child.

602 **Narrative / Reflective.**

- 603 14. Write a letter from a tree to the person who planted it fifty years ago.
- 604 15. Write a short scene where two strangers find common ground while stuck in an elevator.
- 605 16. Write a reflection on what you would tell your future self in a time capsule.
- 606 17. Describe a moment of unexpected kindness you witnessed or imagined.

607 **Professional / Functional.**

- 608 18. Write a professional email declining a job offer while maintaining a positive relationship.
- 609 19. Write a recipe for 'making new friends' formatted as a cooking recipe with ingredients and steps.
- 610 20. Write a product launch announcement for a new renewable energy startup called 'Solara'.